

CS & IT ENGINEERING



'C' Programming

Pointers & Arrays

Lecture No.- 01



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Recap of Previous Lecture



- Fundamentals of Computer, Languages, Translators
- 'C' - history, structure of 'C' Prg
- 'C' Tokens → keywords, Identifiers, Constants, Operators, Strings, Separators
- Data Types
- I/O statements - Console I/O - Printf(), scanf()
- Control Statements - if, if-else, switch, do-while, while, for, break, goto, continue.

Topics to be Covered



- H/w Problems Solving

- Pointers





Topic : Practice Session - 1



#Q. How many lines of output does the following C code produce?

```
float i=2.0;
float j=1.0;
float sum = 0.0;
main(){
    while (i/j > 0.001) {
        j+=j;
        sum=sum+(i/j);
        printf("%f\n", sum);
    }
}
```

A. 8

B. 10

C. 9

☒ D. 11

$$i/j = \frac{2.0}{1.0} = 2.0$$

$$j = 2.0$$

$$\text{sum} = 2.0$$

Print

(1)

$$\frac{2.0}{2.0} = 1.0 > 0.001$$

$$j = 4.0$$

$$\text{sum} = 3.0$$

Print (1)

$$\frac{2.0}{4.0} = \frac{1}{2}$$

$$j = 8.0$$

Print sum
(2)

$$\frac{2.0}{8.0} = \frac{1}{4}$$

$$j = 16.0$$

Print sum
(3)

$$\frac{1}{8}$$

Print
(4)

$$\frac{1}{16}$$

Print
(5)

$$\frac{1}{32}$$

Print
(6)

$$\frac{1}{64}$$

Print
(7)

$$\frac{1}{128}$$

Print
(8)

$$\frac{1}{256}$$

Print
(9)

$$\frac{1}{512}$$

Print
(10)

$$j = 2048.0$$

$$\frac{2.0}{2048.0} > \frac{1}{1024}$$

$$\frac{1}{1024} > \frac{1}{1000}$$

False



#Q. Consider the following C program, It produces _____

```
main()
{
float sum= 0.0, j=1.0,i=2.0;
while(i/j>0.001){
    j=j+1;
    sum=sum+i/j;
    printf("%f/n", sum);
}
}
```

$$\frac{i}{j} > \frac{1}{1000} \text{ True upto } j=1024.0 \Rightarrow \frac{i}{j} = \frac{2.0}{1024.0} = \frac{1}{512}, j=1025, 1026, \dots$$

$j = 1.0, 2.0, 3.0, \dots, 1024.0$
1024 Times True

- A. 0 – 9 lines of output
- B. 10 – 19 lines out output
- C. 20 – 29 lines of output
- ☒ D. More than 29 lines of output



#Q. What does the following program do when the input is unsigned 16 bit integer?

```
#include<stdio.h>
```

```
main(){
```

```
    unsigned int num;    0011 1111
```

```
    int i;
```

```
    scanf("%u", &num);
```

```
    for(i=0;i<16;i++){
```

```
        printf("%d", (num<<i&1<<15)?1:0);
```

```
    }
```

```
}
```

A. It prints all even bits from num

B. It prints all odd bits from num

☒ C. It prints binary equivalent of num

D. None of above

Let num = 63

$1 \ll 15 \Rightarrow$

$$\begin{array}{r} 1\ 0000\ 0000\ 0000\ 0000 \\ 0\ 0000\ 0000\ 0000\ 1111 \\ \hline 0000\ 0000\ 0000\ 0000 = 0 \end{array}$$

op: 00000000 00111111

$i=0$

$63 = 0000\ 0000\ 0011\ 1111$

$i=1$

$63 \ll 1 \Rightarrow 0000\ 0000\ 011\ 11110$

$$\begin{array}{r} 1 \ll 15 = 1000\ 0000\ 0000\ 0000 \\ 0000\ 0000\ 0000\ 0000 = 0 \end{array}$$

$i=2$

0

$i=3$

0

$\vdots$

$i=9$

0

$i=10$
 $63 \ll 10$

$$\begin{array}{r} 11111\ 0000\ 0000\ 0000 \\ 10000\ 0000\ 0000\ 0000 \\ \hline 10000\ 0000\ 0000\ 0000 \\ \text{Non-zero} \end{array}$$

$i=11 = 1$

$i=12 = 1$

$i=13 = 1$

$i=14 = 1$

$i=15 = 1$



Topic : Practice Session - 1



#Q. Consider the following C function definition

```
int f (int x, int y) {
    for (int i = 0; i < y; i++) {
        x = x + x + y;
    }
    return x;
}
```

$$i=0 \Rightarrow x = 2x + y \Rightarrow 2^1 x + 1 \cdot y$$

$$i=1 \Rightarrow x = 2(2x + y) + y \\ = 2^2 x + 2y + y \\ = 2^2 x + 3y$$

$$i=2 \Rightarrow x = 2(2^2 x + 3y) + y \\ = 2^3 x + 7y$$

$$\begin{matrix} 0-1 & 2(0+1)-1 \\ 1-3 & 2(1+1)-1 \\ 2-7 & 2(2+1)-1 \\ 3-15 & 2(3+1)-1 \end{matrix} \left\{ \begin{matrix} 2^{(i+1)} \\ -1 \end{matrix} \right. = k$$

$$\Rightarrow i = y-1 \Rightarrow 2^{(y-1)+1} x + k y = 2^y \cdot x + k \cdot y$$

$$a) x=20, y=10 \Rightarrow 2^{10} \cdot 20 + k(10) > 2^{20} \text{ False}$$

$$b) x=20, y=20 \Rightarrow 2^{20} \cdot 20 + k(20) > 2^{20} \text{ True}$$

$$c) x=20, y=10 \Rightarrow 2^{10} \cdot (20) + k(10) < 2^{10} \text{ False}$$

$$d) x=10, y=20 \Rightarrow 2^{20} \cdot (10) + k(20) > 2^{20} \text{ True}$$

Which of the following statements is/are TRUE about the above function?

- (A) If the inputs are $x=20$, $y=10$, then the return value is greater than 2^{20} *False*
- (B) If the inputs are $x=20$, $y=20$, then the return value is greater than 2^{20} *True*
- (C) If the inputs are $x=20$, $y=10$, then the return value is less than 2^{10} *False*
- (D) If the inputs are $x=10$, $y=20$, then the return value is greater than 2^{20} *True*

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Types of Pointers

① Integer / character / float / double Pointer

② Void Pointer

③ NULL Pointer

④ Array Pointer

⑤ Structure Pointer

⑥ Union Pointer

* ⑦ File Pointer $\Rightarrow$ FILE *P;

⑧ Function Pointer

⑨ Pointer to Pointer

out of Syllabus

⑩ Dangling Pointer

⑪ Near Pointer

⑫ far Pointer

⑬ Huge Pointer

⋮

outdated



2 mins Summary



- H/w Q's Solving
- Pointer Introduction
- Types of Pointers

To be Contd . . .





THANK - YOU